

# John Okyere

## Software Engineer

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### PROFESSIONAL SUMMARY

ICT Education student and Full-Stack Engineer with 3+ years of experience building technology-driven learning solutions. Passionate about transforming education through technology, with hands-on teaching experience at senior high school level and a track record of training 500+ developers and students across Ghana and internationally. Published researcher in financial regulation and digital identity systems. British Council scholar with cross-continental experience across Africa, Europe, and the Americas, committed to innovative, culturally responsive pedagogy.

### EDUCATION

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| <b>BSc ICT Education</b> , <i>University of Education, Winneba</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 01/2023 – Present<br>Winneba, Ghana  |
| <ul style="list-style-type: none"><li>• Pursuing a degree that integrates computing, pedagogy, and curriculum development, with practical teaching experience at senior high school level.</li><li>• Selected as a British Council scholar for the startup innovation programme, gaining exposure to global entrepreneurship and innovation methodology.</li><li>• Served as Head of Media &amp; Publicity for the student association, leading communications, branding, and digital outreach.</li><li>• Organised tech workshops and training sessions equipping students with practical skills in emerging technologies.</li></ul> |                                      |
| <b>Diploma in Blockchain Studies</b> , <i>Althash University</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 06/2023 – 06/2024<br>California, USA |
| Specialized in smart contracts, zero-knowledge systems, and decentralized infrastructure, complementing hands-on experience in blockchain-based verification systems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                      |

### PROFESSIONAL EXPERIENCE

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| <b>Teaching Intern</b> , <i>Toase SHS</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 10/2025 – 05/2026<br>Kumasi, Ghana |
| <ul style="list-style-type: none"><li>• Planned and delivered structured lessons in ICT, Elective ICT, and introductory programming to senior high school students, adapting content to diverse learning needs and ability levels.</li><li>• Designed interactive, hands-on classroom activities that improved student engagement and confidence in digital literacy and core computing concepts.</li><li>• Mentored final-year students on practical problem-solving and academic preparation, contributing to measurable improvements in classroom participation.</li><li>• Applied culturally responsive teaching strategies to make abstract programming concepts accessible and relevant to students' everyday experiences.</li><li>• Collaborated with subject teachers to align lesson delivery with the national ICT curriculum framework.</li></ul> |                                    |

**Technical Lead, Kayaba Labs** 

09/2025 – Present  
Accra, Ghana

- Designed and delivered structured technical training programmes for 30+ developers in web development, AI, and blockchain, achieving an 85% skill certification rate.
- Built a full-stack Learning Management System (LMS) using Next.js, Node.js, and PostgreSQL, scaling to 500+ active learners within 4 months of launch.
- Mentored junior developers through project-based learning, resulting in 20+ deployed real-world projects.
- Managed 5+ concurrent engineering projects while maintaining a teaching and mentorship role within the organisation.

**Resident Fellow, Ethereum Ethiopia**

01/2025 – 02/2025  
Addis Ababa, Ethiopia

- Selected from 200+ global applicants for competitive Ethereum builder fellowship, delivering 3 technical milestones, including an AI-powered evidence verification system and EVM-based carbon credit smart contracts over 6 weeks.
- Led full-stack development for EcoFusion carbon credit platform, implementing AI verification with 90%+ accuracy and tokenizing 500+ sustainability actions on Hedera DLT during a 6-week build sprint.
- Achieved top 5 ranking among 40+ fellowship projects by implementing a multi-modal AI verification pipeline on Hedera DLT that validated carbon credit claims with 90%+ accuracy recognized by judges for combining AI and on-chain trust mechanisms.

**Founding Engineer, CredAxis**

02/2025 – 12/2025  
Remote

- Built and launched decentralized credential verification platform from concept to live beta in 3 months, serving 200+ users with sub-3-second on-chain verification and 100% tamper-proof record integrity.
- Architected full-stack platform using Next.js, Node.js, and Solidity, deploying smart contracts that processed 500+ verification requests and achieving sub-100ms frontend response times through optimized UX flows.

**AOC DevConnect Scholar, DevConnect Argentina (EF)**

09/2025 – 10/2025  
Buenos Aires, Argentina

- Secured \$1,000 competitive scholarship (1 of 20 African recipients) to represent African builders at Devconnect Argentina, establishing 8+ partnerships with Ethereum core teams that led to ongoing collaboration on the Base Carbon Meter DePIN project.
- Collaborated with 50+ Ethereum and Celo core developers across 10+ technical workshops on ZK cryptography and DeFi infrastructure, applying learnings to architect the ZK verification layer for Base Carbon Meter and the ZK-proof components of CredAxis.
- Gained applied expertise in ZK-SNARKs and proof systems, architecting a ZK verification layer for Base Carbon Meter that processed energy telemetry with 99.8% tamper-proof validation accuracy, reducing fraudulent data risk by an estimated 40%.

**ZK Scholar & Resident Fellow, Dev3Pack Global**

09/2025 – 10/2025  
Palermo Soho,  
Argentina

- Completed competitive ZK Scholar fellowship, mastering Groth16 proof systems and ZK-SNARKs, then immediately deploying a production ZK verification layer that processed 50GB+ of live energy telemetry data with 99.8% validation accuracy for Base Carbon Meter.
- Joined Base Carbon Meter DePIN project through fellowship, contributing full backend infrastructure and ZK verification layer that enabled tokenization of 1,000+ verified renewable energy offsets on Base.

- Built backend AWS infrastructure using S3, Lambda, and DynamoDB, processing 50GB+ of energy telemetry data with 99.8% validation accuracy through a custom ZK verification layer integrated with Amazon Bedrock LLM routing.

**ICP Ambassador, ICP Hubs Global**

01/2025 – 12/2025  
Central Region, Ghana

- Designed and facilitated 5+ developer education workshops reaching 600+ participants, creating localised learning materials on blockchain and decentralised applications.
- Mentored 200+ students and early-career developers on building solutions using Motoko and Rust, resulting in 40+ prototype deployments.
- Developed curriculum-style content translating complex technical concepts into accessible learning formats for diverse audiences.
- Grew community developer participation by 40% through consistent educational engagement and peer learning initiatives.

**CyberSecurity Intern, Pinnacle Labs**

06/2024 – 08/2024  
Bengaluru, India

- Built password strength analyzer evaluating 10,000+ passwords with 94% vulnerability detection accuracy, implementing AES-256, DES, and RSA encryption algorithms to harden data protection across 3 internal systems.
- Developed keylogger simulation for security research, identifying 6 critical input vulnerability patterns and deploying 4 defensive countermeasures that reduced system attack surface by 35% across 3 internal environments.
- Developed 3 internal security tools and training materials, improving cybersecurity practices for a 15-person engineering team and reducing simulated incident response time by 30%.

**Machine Learning Intern, Trestle Academy (4th-IR) **

08/2024 – 09/2024

- Developed and deployed a CNN-based pneumonia detection model using TensorFlow and Keras, achieving 95% accuracy on thousands of X-ray images matching radiologist-level diagnostic precision.
- Enhanced model performance through data preprocessing, augmentation, and automated hyperparameter tuning, resulting in a 22% accuracy boost, 20% reduction in overfitting, and 18% faster training.
- Deployed the model with FastAPI for real-time inference, reducing diagnosis time by 20% and improving early detection rates by 15%.

**Cloud Engineer Intern, AmaliTech Services GmBH **

06/2024 – 08/2024  
Remote

- Architected and deployed scalable cloud infrastructures using 30+ AWS services (including EC2, RDS, and S3), boosting system performance by 25% and cutting operational costs by 15%.
- Built and optimized a data pipeline that processed 2TB of Netflix data from Amazon S3, leveraging Amazon QuickSight for analytics and improving content recommendation accuracy by 20%.
- Implemented CI/CD automation with GitHub Actions and AWS, enabling 5 deployments per week, reducing deployment time by 75%, achieving 99.9% system uptime and 80% faster page load speeds.

## TECHNICAL SKILLS

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### Backend & APIs

- Node.js, Bun, Express.js, Hono, ElysiaJS, REST APIs, WebSockets, Zod, Redis / Upstash, FastAPI, Rust

### Product & Design

- Figma, design systems, product thinking, user testing

### Tooling & Infrastructure

- Git, GitHub Actions (CI/CD), Vercel, Docker, Turborepo, AWS, GCP, Azure, EC2, S3, Render, Railway

### Education & Pedagogy

- Lesson planning, curriculum design, culturally responsive teaching, project-based learning, LMS development

## Publications

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### **From Prohibition to Supervised Perimeter: How West African Jurisdictions Are Domesticating FATF Recommendation,** *Journal of Financial Regulation, Oxford*

Conducted a comparative analysis of virtual asset regulation in Ghana, Nigeria, and South Africa, examining how each implements Financial Action Task Force Recommendation 15 through distinct legal frameworks. Identified key differences in regulatory design, enforcement capacity, and market impact, highlighting supervision capacity not legislation as the critical factor in effective AML/CFT compliance.

### **Regulating Virtual Asset Service Providers in Ghana: A Case Study of the Virtual Asset Service Providers Act, 2025,** *Journal of Banking Regulation, Oxford*

Analyzed Ghana's Virtual Asset Service Providers Act, 2025 (Act 1154), examining its regulatory framework for VASPs, including licensing, AML/CFT compliance, and multi-agency oversight. Found strong alignment with Financial Action Task Force standards, while highlighting implementation challenges in supervision, cross-border activity, and compliance burden

### **ZKP-IDFS: A Zero-Knowledge Proof-Based Digital Identity Framework for Financial Inclusion in Sub-Saharan Africa** [↗](#), *IEEE ACCESS*

Designed **ZKP-IDFS**, a privacy-preserving digital identity framework using zk-SNARKs, verifiable credentials, and USSD/SMS channels to enable secure KYC for low-connectivity environments. Achieved sub-5s verification on 3G and low-cost on-chain validation, demonstrating alignment with Financial Action Task Force guidelines and African data protection laws.

## Certificates

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- AWS Cloud Professional [↗](#)
- Registered Blockchain Professional (ICORBP) [↗](#)

## References

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**Prof Delali Kwasi Dake**, *Head of Department, Dept of ICT Education*, University of Education, Winneba  
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**Ms. Esther Gyimah**, *Senior Lecturer, Dept of ICT Education.*, University of Education, Winneba  
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